Adrian Rodrigues

Senior Graphics / Rendering Programmer

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SUMMARY

Senior Graphics Programmer with 4+ years in AAA on Unreal Engine 4 and 5 (PC, PS5, Xbox Series X|S), specialized in rendering performance, memory and stability. Builds Cerberus, a personal C++/Vulkan path tracer.

PROFESSIONAL EXPERIENCE

Virtuos — Jun 2022 – Present

Senior Graphics Programmer · *Third Kind Games* — *Feb 2025 – Present*

Unannounced UE5 project (under NDA) — *Jul 2026 – Present*

- Reduced Virtual Shadow Map cost for skinned foliage by **~29%** on PS5 by moving its invalidation to the static path and time-slicing GPU invalidation across frames when lighting changes.
- Profiled the Nanite Visibility pass, identifying voxels (**~28%**) and programmable raster (**~20%**) as its largest costs; built tools to switch foliage Nanite settings to voxels and tuned programmable raster, density and rasterizer configuration, improving the pass by **~20–25%**.
- Broke down Base Pass shading with A/B tests, measuring **~7–9%** savings each from derivative-heavy shading and vertex-side material features, and turned all findings into a prioritized action plan for Art and Tech Art.
- Reduced Virtual Texture upload spikes by **~30–40%**.

Mavrix · *Unreal Engine 5, C++, HLSL, DirectX 12* — *Feb 2025 – Jul 2026*

- Worked extensively on memory within tight, fixed console budgets (notably Xbox Series S): fixed multiple memory leaks, improved memory management in Errant (custom plugin) and used memory profiling tools to drive game-wide optimizations.
- Optimized Virtual Shadow Maps with a distance-based ray-count falloff for Shadow Map Ray Tracing and custom cache invalidation for large, dynamically streamed environments.
- Backported several Unreal Engine fixes and improvements to resolve memory issues, improve performance and increase stability.

Senior Graphics Programmer · *Black Shamrock* — *Aug 2023 – Feb 2025*

Judas, Unreal Engine 4 · promoted from Graphics Programmer in Nov 2024

- Developed a render-state streaming system that reduced the primitives processed per frame and improved render-thread efficiency; backported UE5 compute visibility to UE4.
- Made runtime reflection-probe generation time-sliced and on-demand for procedurally generated (PCG) content; mentored a junior developer.

Junior Graphics Programmer · *Virtuos Paris, Judas (UE4)* — *Jun 2022 – Aug 2023*

- Culling to reduce draw calls and lighting optimization within performance budgets.

SELECTED PROJECTS

- **Cerberus (C++, Vulkan)** — personal renderer with a hardware ray-traced path tracer (Vulkan KHR ray tracing, bindless textures) and an A-SVGF denoiser for real-time 1-spp output.
- Render graph with automatic barrier derivation and Dynamic Rendering scope merging, over a thin RHI with typed command lists and handle-based resource ownership.

SKILLS

Rendering (UE4–5)	Nanite, Virtual Shadow Maps, Lumen and ray tracing, (Runtime) Virtual Textures, culling and LOD, foliage, DLSS/FSR 4 integration
Languages and APIs	C++, C#, HLSL, GLSL · DirectX 12, Vulkan, OpenGL · PC, PS5, Xbox Series X|S
Profiling and tools	Unreal Insights, RenderDoc, PIX, Razor, Git, Perforce

EDUCATION

MSc Computer Science, Instituto Superior Técnico *(2019 – 2021)* · **BSc Computer Science**, University of Lisbon *(2016 – 2019)*